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University of Glasgow



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Joseph Henry Briggs

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Declaration

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Abstract

Chapter 1

Gravitational Wave Interferometers

1.1 Introduction

The aim of this chapter is to outline some things about our universe we would like to know, how a gravitation wave allows this to be known, how an interferometer can be used to measure gravitational waves, the fundamental noise sources these detectors face. It aims to relate simple general-relativity based arguments to propose a gravitational wave detector. These noise sources can be reduced by making changes to the readout of the LIGO detector and by building new interferometers that use new technology. In this thesis, power of the laser and BHD is explored. Third generation detectors may operate at 2um, and so 2um detectors were characterised.

1.2 Sources of GW

order = physics of GW generation -> detected sources -> undetected sources. Try to emphasise the need to improve current detectors to detect the the undetected

sources? Gravitational waves are emitted from mass quadrupole systems ¹. As the coupling between mass and the curvature of space-time is weak, gravitational waves are only observable in systems that have large masses involved.

To date, gravitational waves emanating BBH, BNS, and, possibly, from a binary consisting of a black hole and a neutron star (NSBH) [1,2] have been detected. The waveform of gravitational radiation emitted from these systems is known as a chirp signal. These signals increase in frequency as the two objects get closer to merging. Due to the geometry of these systems, the frequency of the gravitational waves are in the audioband with the cut-off frequency, determined by the radius of the objects that will merge, between 100 Hz-10 kHz. A typical gravitational wave strain amplitude of these types of systems, all of which have a total mass of $10M_{\odot} - 100M_{\odot}$ and are of the order 100 Mpc to 10 Gpc away from Earth, is $h \sim 10^{-21}$.

Other sources of gravitational waves include pulsars and magnetars. Asymmetries in these systems will cause them to emit a gravitational wave with a constant frequency. Burst signals will be emitted by supernovae. There will be a stochastic background of gravitational waves, similar to the cosmic microwave background radiation, as well as a background level of gravitational waves due to the merging of compact binaries. Dark matter and Cosmic strings may give rise to gravitational waves. However, the gravitational waves from these systems are yet to be directly detected.

¹Due to the conservation of mass/energy, there is no mass monopole radiation. Due to the conservation of momentum and angular momentum, there is no dipole radiation. The next term that can generate radiation is the quadrupole moment $h_{ij} = \frac{1}{R} \left(\frac{2G}{c^4} \ddot{I}_{ij} \right)$.

Observation Run	Dates	Detections	Detectors
O1	12/09/2015 - 19/01/2016	3	aLIGO
O2	30/11/2016 - 25/08/2017	8	aLIGO, Virgo
O3a	01/04/2019 - 01/10/2019	39	aLIGO, Virgo
O3b	01/11/2019 - 04/30/2020	23	aLIGO, Virgo
O4	2021	-	aLIGO, Virgo, KAGRA
O5	date?	-	LIGO A+, Advanced Virgo, KAGRA, LIGO India

Table 1.1: table of observation runs

1.3 Astrophysics and Cosmology observation runs and discoveries (detector hierarchy)

The need for increasing detections... Bayesian analysis requires evidence from multiple detections to inform prior distributions. In order to get better statistical models of BBH and BNS populations, multiple detections are required. To detect continuous wave signals, the signal to noise ratio improves with the total observing time of the detector. With longer observation times, it is more likely that an exotic event will be measured. For these reasons, the gravitational wave detectors are operated for year long time-scales with minimal interruption. These are known as observation runs. Between observation runs, the detectors undergo hardware upgrades. The current observation runs are summed up in table 1.1.

1.4 Measuring Gravitational Waves with an Interferometer

general relativity (BIG L shaped IFO) Einstein's general theory of relativity accurately describes the gravitational waves measured with the LIGO and Virgo gravitational wave detectors [3]. Gravitational waves are propagating longitudinal perturbations in the curvature of space-time, and can be split into a basis set containing two polarisations, plus and cross (reference gravitation).²

The effect these waves have, in the local Lorentz frame, on test masses on Earth is to move them by a small amount. To detect plus polarised gravitational waves, the end mirrors of a Michelson interferometer can be used as the test masses. In a Michelson interferometer, light is split with a beam splitter and directed towards mirrors. These mirrors reflect the light back to the beam splitter where it recombines. When the mirrors at the end of a Michelson interferometer are moved differentially by a plus polarised gravitational wave. This causes a phase modulation in the light in the arms, and the intensity of the light at the *anti-symmetric* port is modulated

²In the local Lorentz frame of the beam splitter, for any feasible Earth based detector, the effect of a gravitational wave on the wavelength of the light is negligible compared to its effect on the positions of the test masses. This is due to the strong equivalence principle. This states that 'All physical laws that are expressible as a tensor equation in special relativity have the exact same form in a local Lorentz frame of a curved space time'. In the local Lorentz frame of a gravitational wave detector, the metric has the form $g_{\alpha\beta} = \eta_{\alpha\beta} + \mathcal{O}(h)$, where $\eta_{\alpha\beta}$ is the Minkowski space-time metric and $\mathcal{O}(h)$ is a perturbation of the order $\frac{2\pi L}{\lambda_{\text{EM}}}^2 h$. The Minkowski space-time metric is flat, so in the local Lorentz frame the metric is, essentially, flat and the gravitational waves cause the test masses to move by $\Delta L = \frac{1}{2}hL$. Maxwell's equations, due to the strong equivalence principle, are modified by a term that is of the order $\frac{\delta\lambda_{\text{EM}}}{\lambda_{\text{EM}}} \sim \frac{2\pi L}{\lambda_{\text{EM}}}^2 h$, and so if $\frac{2\pi L}{\lambda_{\text{EM}}}$ is sufficiently small, as it would be for an Earth based interferometer, then the effect of the gravitational wave on the light is much smaller than the effect of the gravitational wave on the test masses. Alternatively, in the traceless-transverse gauge, the position of the test masses is fixed and the gravitational wave changes the wavelength of the light.

at the frequency corresponding to the frequency of the phase modulation. By placing a photo detector in this beam, it is possible to measure the differential signal generated by the gravitational wave.

1.4.1 Required Arm Length of the Michelson Interferometer

The optimal arm length of the interferometer is determined by $1/4$ of the wavelength of a gravitational wave. For BBH and BNS, the gravitational wave signal will be at 100 Hz, and so the wavelength of the gravitational wave will be of the order 3000 km. It is not possible to construct such an interferometer on Earth; the sites that were available to construct the current generation of gravitational wave detectors allow for 4 km arms. Note that if the arm were to be made longer, the sensitivity of the detector would decrease as the phase difference accumulated in the arms would no longer result in maximum constructive interference at the antisymmetric port.

To reach the optimal arm length on a 4 km site, the number of times that the light bounces off the test mass, B , needs to be increased. Numerically, B should be ~ 250 . Delay lines and Fabret-Perot cavities within the arms of the interferometer were investigated. Ultimately, due to technical considerations in the geometry of the mirrors required to implement a delay line interferometer, Fabret-Perot cavities were selected as a means of increasing the amount of bounces of the light off of the test masses. This means that B is a statistical quantity, and the average number of bounces will be B . [more detail/less rotating hand theorem?](#)

1.5 Sources of noise in Gravitational wave Interferometers.

The challenge faced when detecting gravitational waves is a result of the gravitational wave strain h causing a minuscule change in the phase accumulated by light in the arm cavities. The phase shift, $\Delta\phi_{\text{EM}}$, acquired by the laser beam of wavelength λ_{EM} when a gravitational wave with strain h and with optimal wavelength for the arm length interacts with the detector is

$$\Delta\phi_{\text{EM}} = \frac{4\pi LB}{\lambda_{\text{EM}}} h. \quad (1.1)$$

The choice of $\lambda_{\text{EM}} = 1064 \text{ nm}$ as the wavelength which the detector operates with is determined by the low loss and low thermal noise coatings and test masses that can be fabricated for 1064 nm light. Numerically, for a gravitational wave of strain 10^{-23} interacting with LIGO, the change in phase shift that the light experiences is $\sim 10^{-11}$.

The required energy of the laser beam E to measure the tiny phase shift that the light experiences due to the gravitational wave strain is determined by the time-energy uncertainty principle $\Delta E \Delta T \geq \hbar/2$. The uncertainty in time is related to the uncertainty in phase by $\Delta\phi = \omega_{\text{EM}} \Delta T$. The energy used to measure this time is determined by half of a the gravitational waves period (if the zero crossings of the EM wave are being used). This time T corresponds to $1/2f$, where f is the frequency of the gravitational wave. The power of the laser is $I_0 = \hbar\omega_{\text{EM}}N$, where N is the number of photons detected in the time interval T , and so the energy used

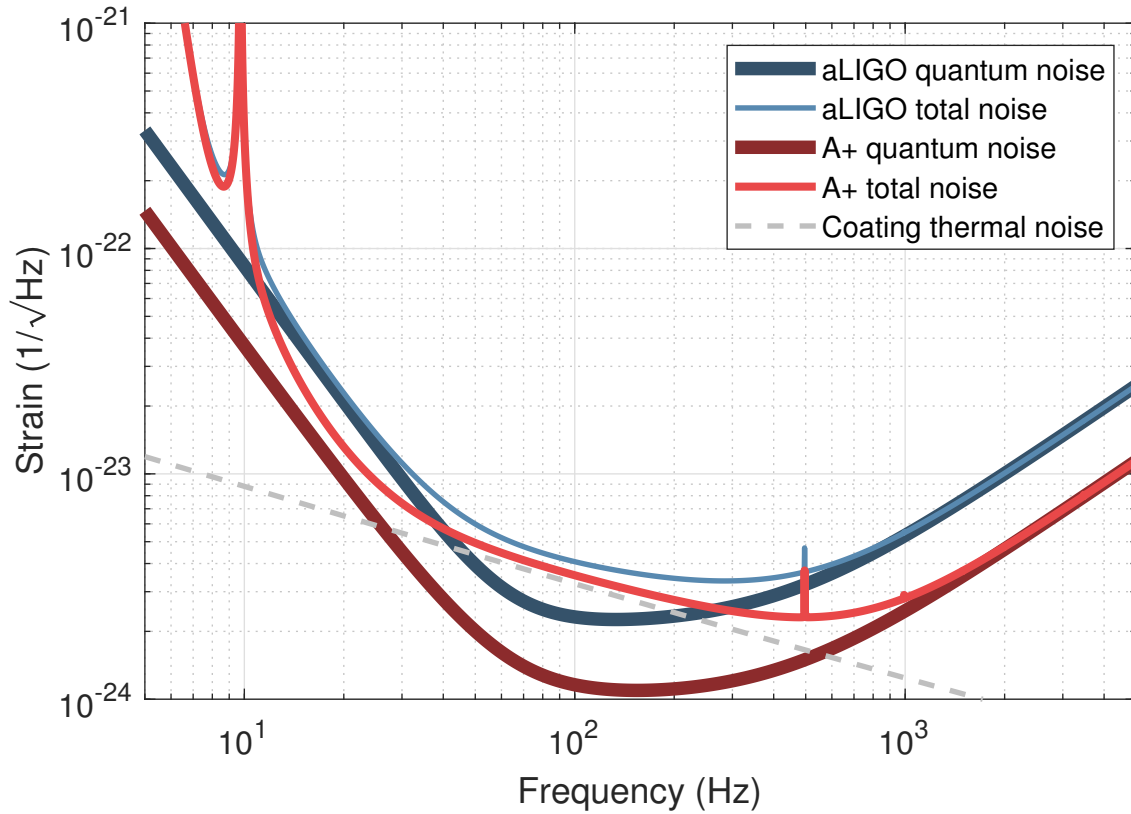


Figure 1.1: The Strain Sensitivity of the aLIGO and A+ detectors. The noise due to radiation pressure and photon shot noise is shown in dark-blue (aLIGO) and dark-red (A+). Above 10 Hz, quantum effects are one of the limiting sources of noise; with the other being the noise due to Brownian motion of the mirror's coating. Below 10 Hz, classical sources of noise, such as seismic motion, limit the sensitivity of both interferometers... figure out what is important to say here!

to measure the time is $E = I_0 T = I_0 \frac{1}{2f} = \hbar \omega_{EM} N$. As the laser is in a coherent state, the uncertainty in the number of photons detected in time T is $\Delta N = \sqrt{N}$. This is known as shot noise. The uncertainty principle may now be written as $\Delta E = \hbar \omega_{EM} \Delta N$ which can be expressed as

$$N \geq \frac{1}{4(\Delta \phi)^2}. \quad (1.2)$$

To beat the photon shot noise by a factor of 10, the required number of photons to measure a gravitational wave with a strain amplitude of 10^{-23} is $\sim 10^{22}$. For a gravitational wave with a frequency of 100 Hz, the laser power required entering the arm cavity would be ~ 1 MW. No such laser exists.

To increase the laser power that is injected into the beam splitter, a power recycling cavity can be used. Gravitational wave detectors are operated on the *dark fringe* meaning that there is a low amount of power at the antisymmetric port and most of the injected laser power gets reflected from the beam splitter towards the laser. This means that the beam splitter acts as a mirror. Drever [?] proposed that a mirror, called the power recycling mirror (PRM), could be placed in the between the laser and the beam splitter and this would form a cavity. By tuning the position of the PRM such that the the light reflected from the interferometer and the light from the laser add in phase, the power would be ‘recycled’ and the total power impinging on the beam splitter would be increased. Current laser sources that can be operated for year-long time scales are capable of producing of the order 100W, and so the power recycling factor needs to be $\sim 10^4$. This technique has been successfully implemented by many large scale interferometers. Does this need expanding,

or just some references to how PRC is implemented e.g. matching transmission to losses.

shot noise and laser power

quantum efficiency

power recycling

signal recycling to shape quantum noise

squeezed light

frequency dependent squeezing

radiation pressure noise?

other topologies of detectors

other noise sources

1.6 The future of gravitational wave detectors

new science that can be done with these detectors? could be in the first section

cyro

silicon

2um or 1550

1.7 conclusion

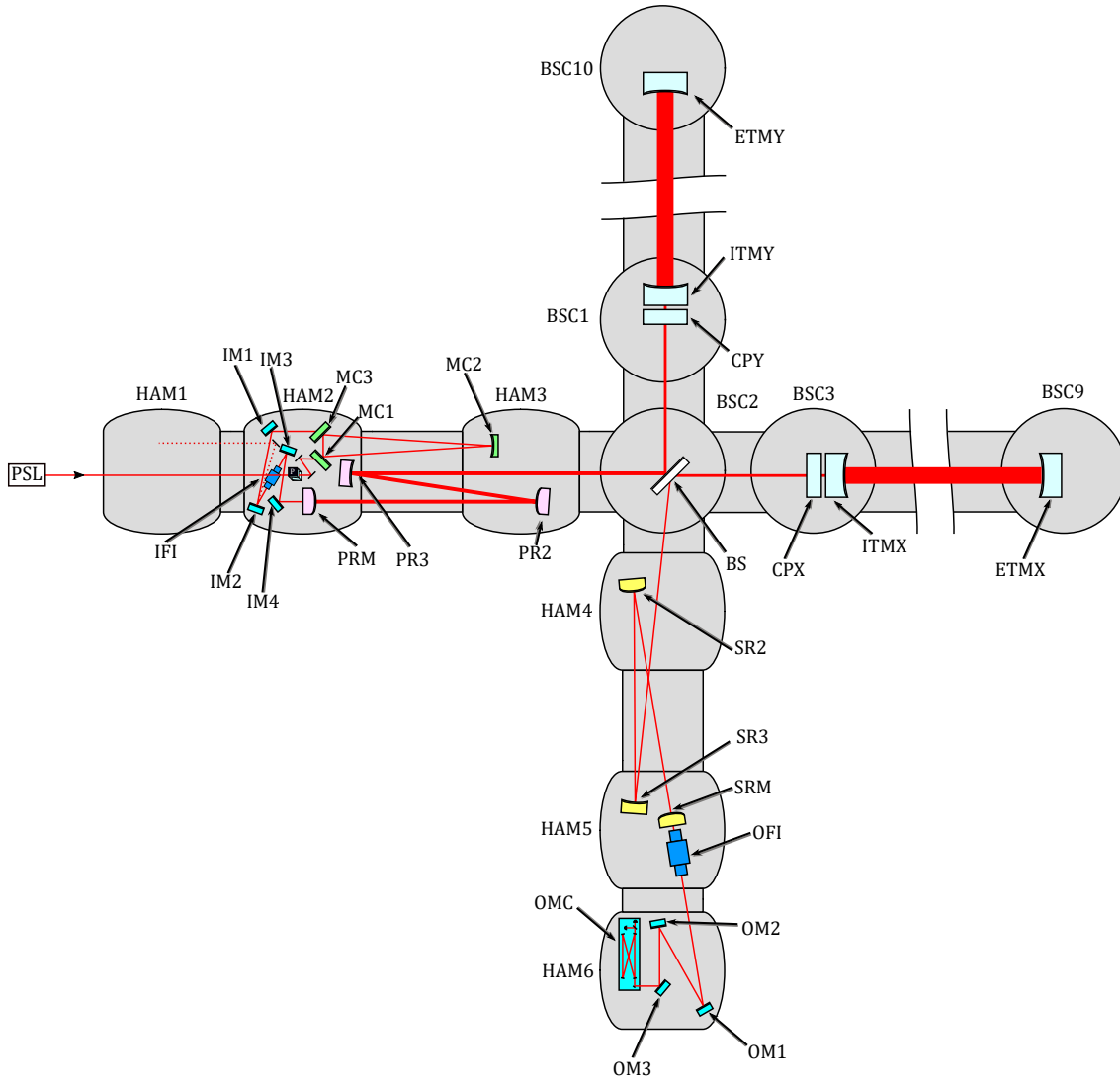


Figure 1.2

Bibliography

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